

Supervised Learning of Behaviors

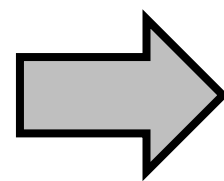
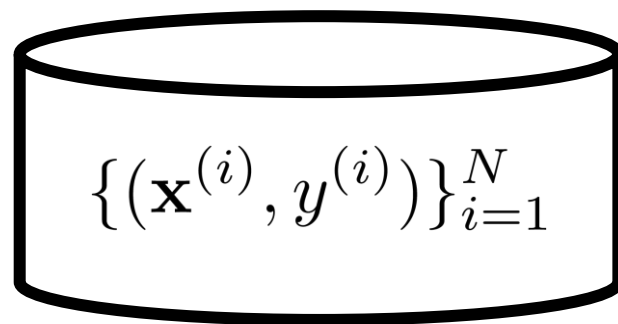
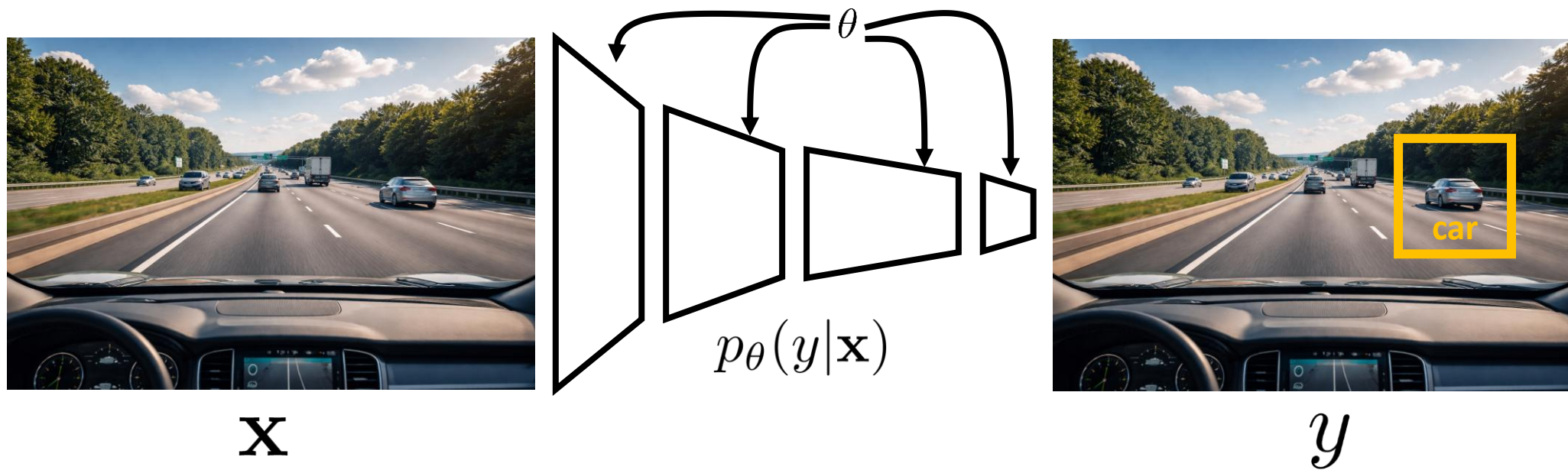
CS 185/285

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UC Berkeley



Part 1:

From supervised learning to imitation learning



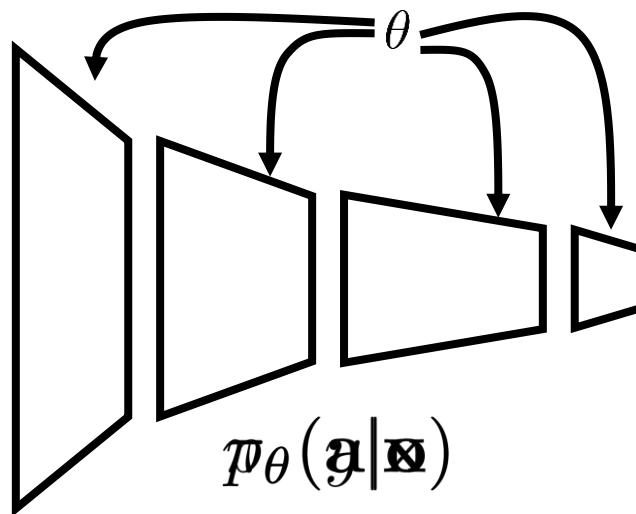
$$\arg \max_{\theta} \sum_{i=1}^N \log p_{\theta}(y|\mathbf{x})$$

maximum likelihood estimation

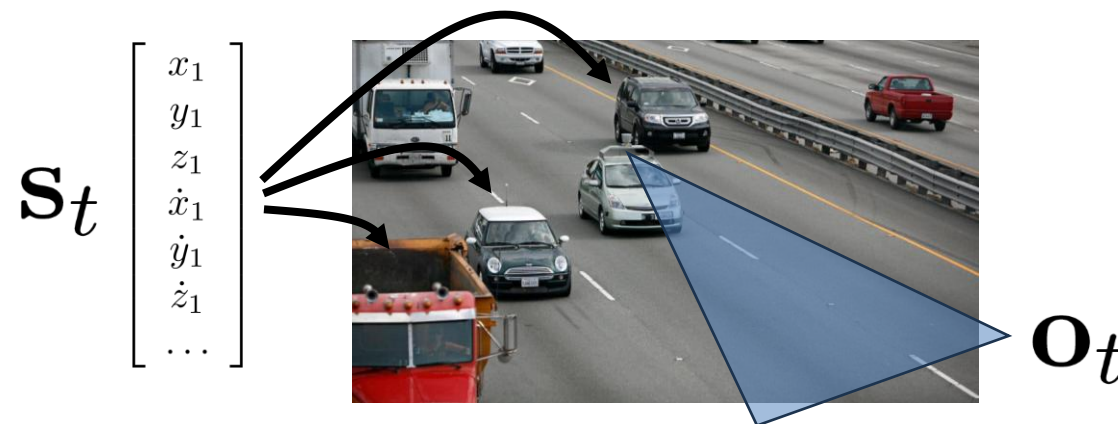
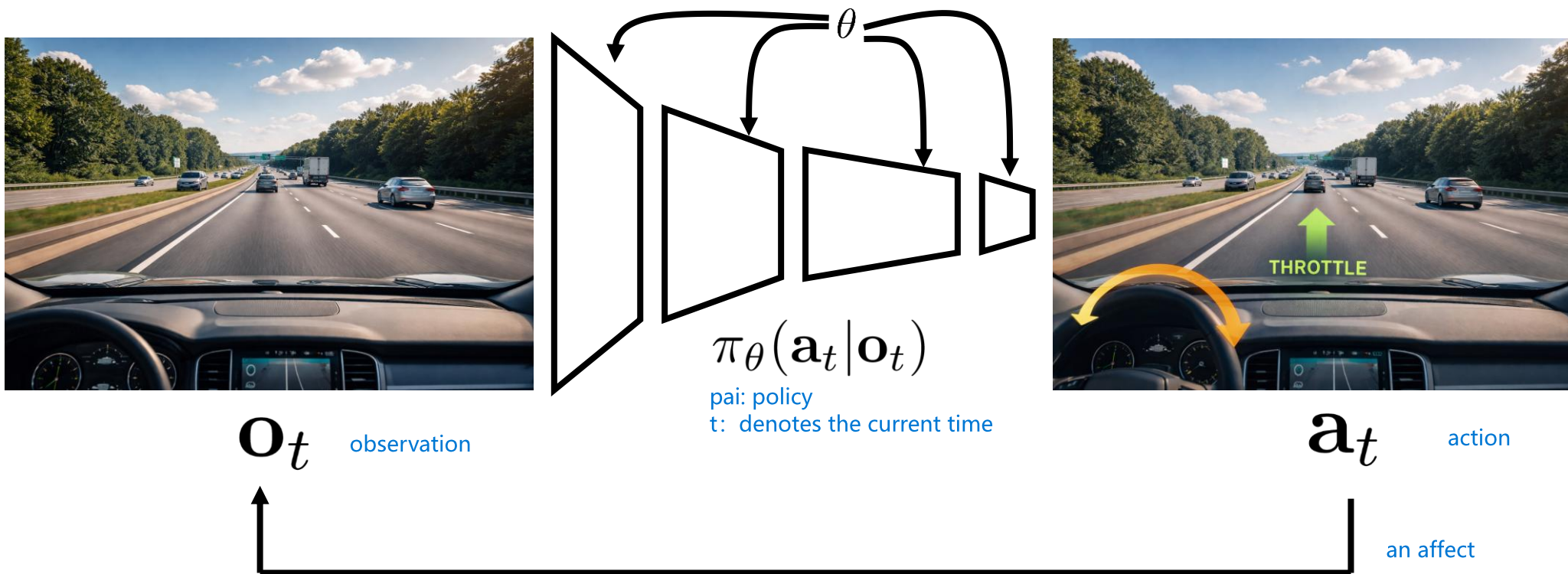
a.k.a. supervised learning



$\mathbf{x}$

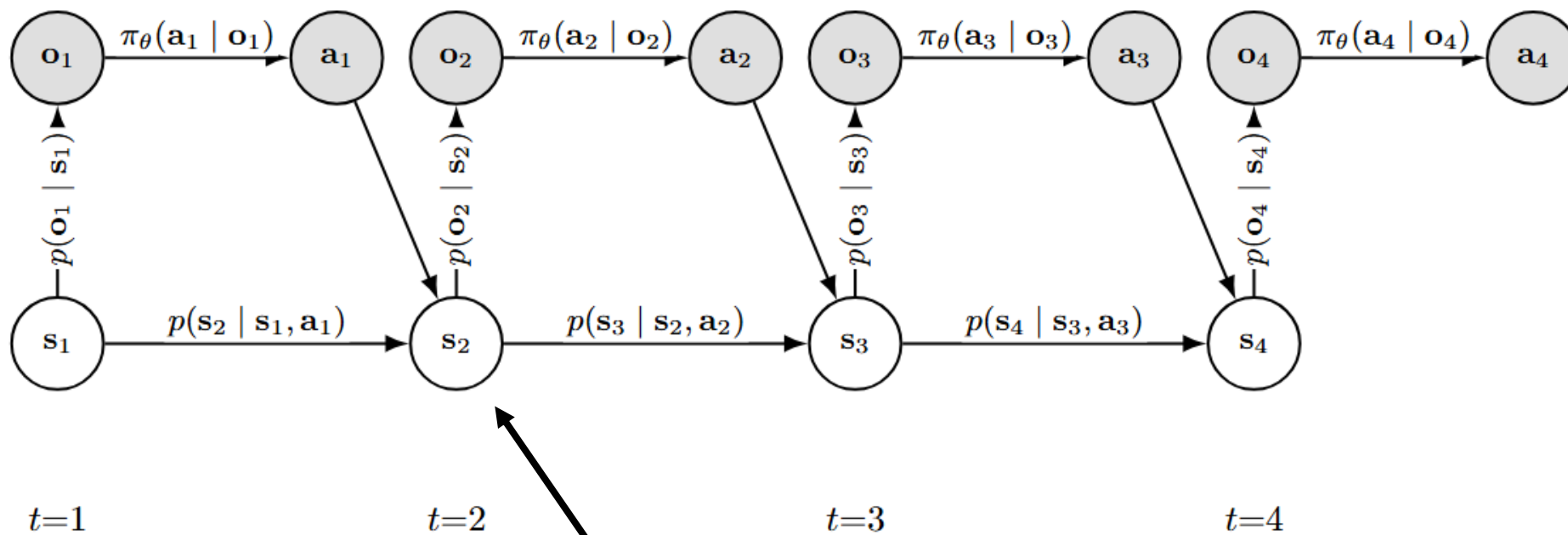


$\mathbf{a}$



$\mathbf{S}_t$: t时刻的state, 区别于 $\mathbf{O}_t$ 是观测值。
 $\mathbf{S}_t$ 是全知的, 当前状态的所有信息都知道

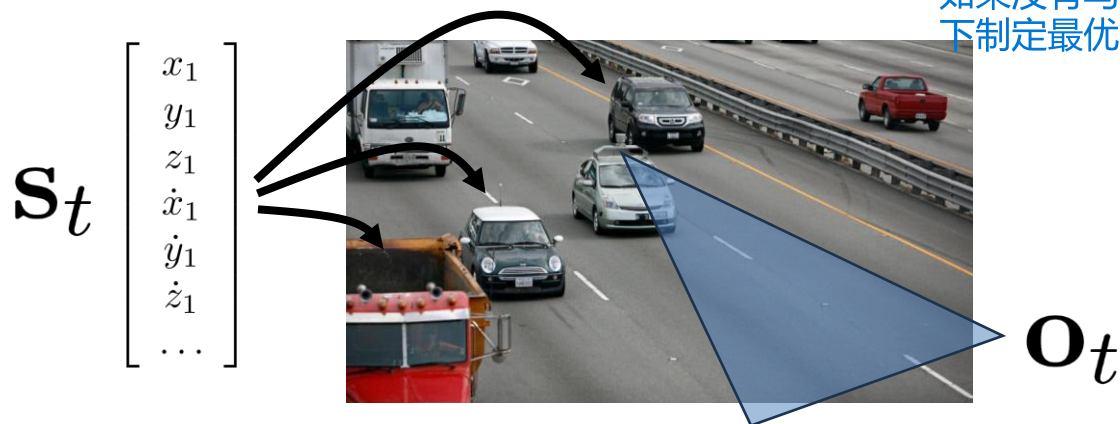
observation可能不足以推断出t时刻正确状态!!!



Markov property: $s_{t+1} \perp s_{t-1} | s_t$

state是具有马尔科夫链性质的， s_{t+1} 只取决于 s_t ，而不取决于之前如何达到 s_t 状态

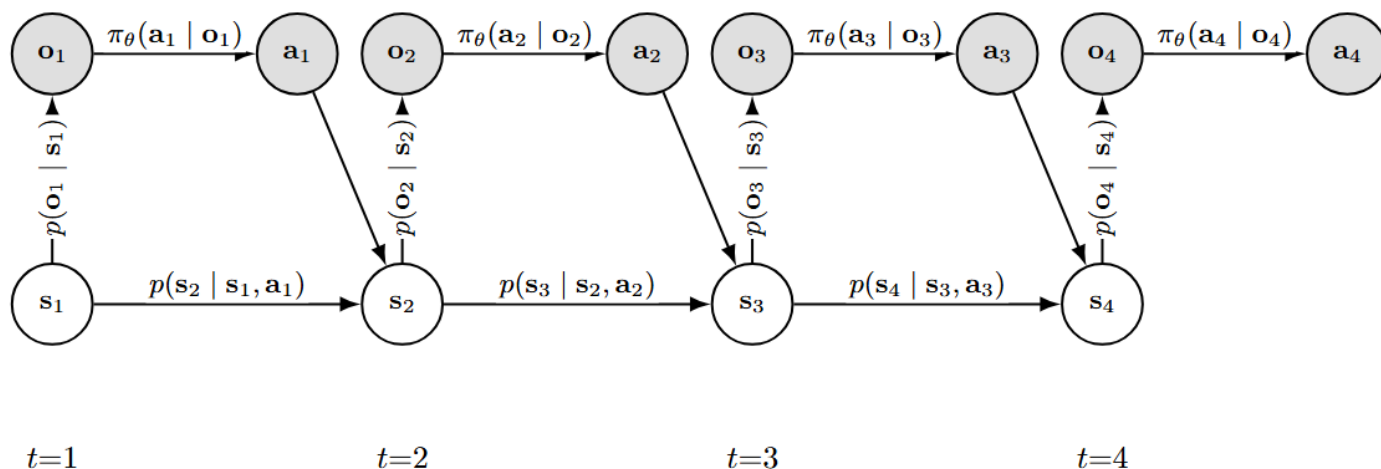
如果没有马尔科夫性质，我们就无法在不完整考虑历史的情况下制定最优策略，马尔科夫性质是RL的基础



Partially observed case

部分可观测马尔可夫决策过程

在无模型（Model-free）强化学习中，智能体通常不需要知道环境的转移概率，它只需要通过不断试错来优化策略 π 。就像开车时只能看到前方视野（有盲区）



$$\pi_\theta(\mathbf{a}_t | \mathbf{o}_t)$$

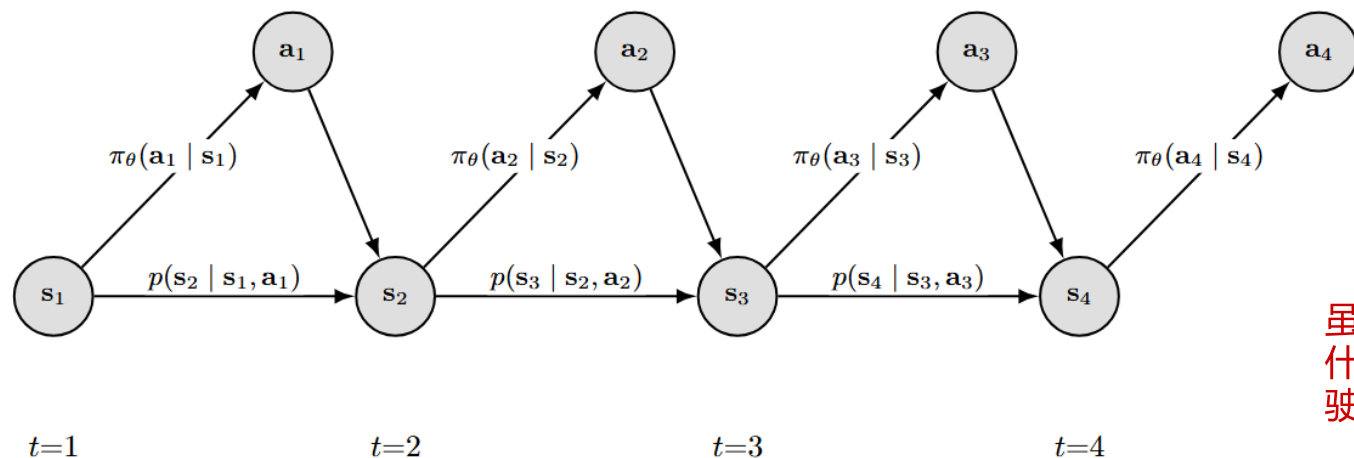
$$p(\mathbf{o}_t | \mathbf{s}_t)$$

$$p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$$

typically
don't need
to know this

完全可观测情况，其实就是标准的MDP过程。就像下象棋，能看到全局棋盘。复杂度低

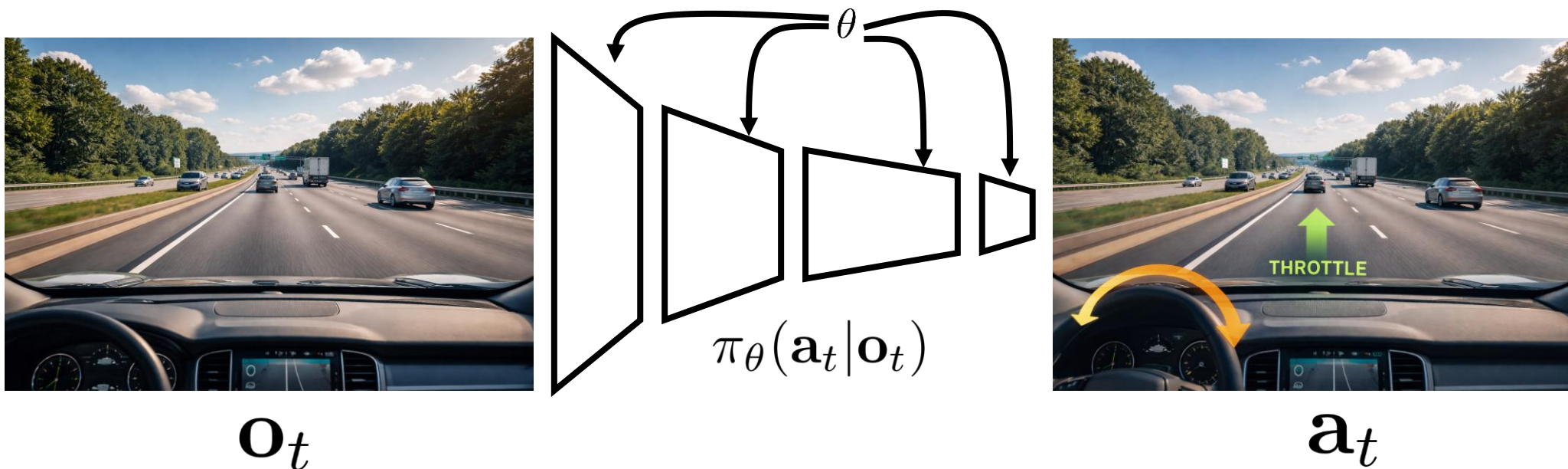
Fully observed case $\mathbf{O}_t = \mathbf{S}_t$



$$\pi_\theta(\mathbf{a}_t | \mathbf{s}_t)$$

$$p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$$

虽然环境的物理运行逻辑（状态转移）是类似的，但智能体能拿到什么信息决定了策略设计的难度。在实际应用（如机器人、自动驾驶）中，我们大多处理的是第一种“部分可观测”的挑战。



Data? $\{(\mathbf{o}_1^{(i)}, \mathbf{a}_1^{(i)}, \dots, \mathbf{o}_H^{(i)}, \mathbf{a}_H^{(i)})\}_{i=1}^N$

“demonstration trajectory”

在强化学习或模仿学习中由专家演示产生的一系列状态和动作的序列

$$\arg \max_{\theta} \sum_{i=1}^N \sum_{t=1}^H \log \pi_{\theta}(\mathbf{a}_t^{(i)} | \mathbf{o}_t^{(i)})$$

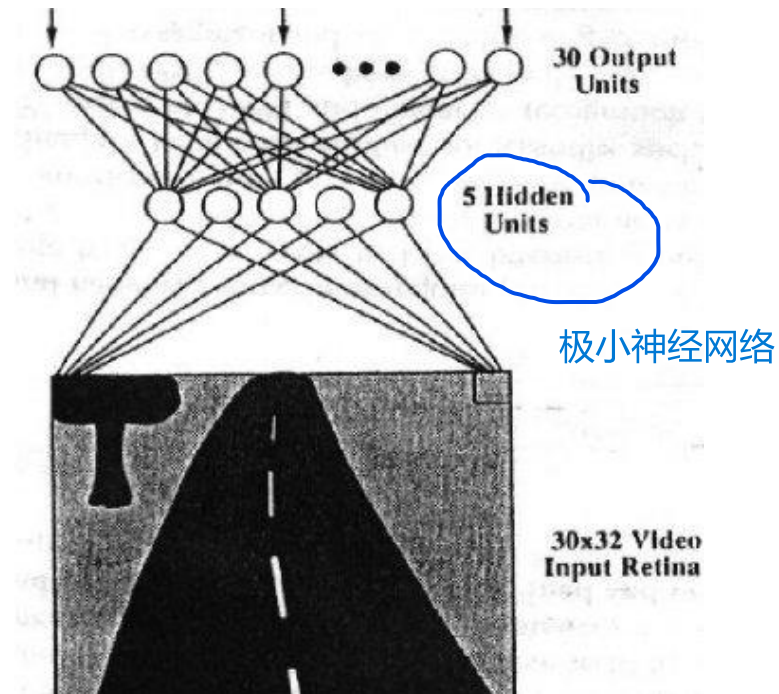
“behavioral cloning”

Part 2:

The behavioral cloning algorithm

ALVINN: Autonomous Land Vehicle In a Neural Network

1989



1. Collect data by asking a person to make demonstrations
here person is called the Expert

$$\{(\mathbf{o}_1^{(i)}, \mathbf{a}_1^{(i)}, \dots, \mathbf{o}_H^{(i)}, \mathbf{a}_H^{(i)})\}_{i=1}^N$$

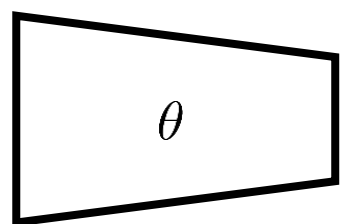
找概率最大的policy, 模仿专家的行为

2. Run supervised learning (a.k.a. maximum likelihood estimation) on this data with your favorite neural network

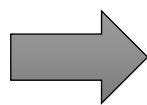
$$\arg \max_{\theta} \sum_{i=1}^N \sum_{t=1}^H \log \pi_{\theta}(\mathbf{a}_t^{(i)} | \mathbf{o}_t^{(i)})$$



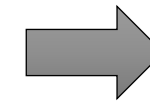
$\mathbf{o}_t$



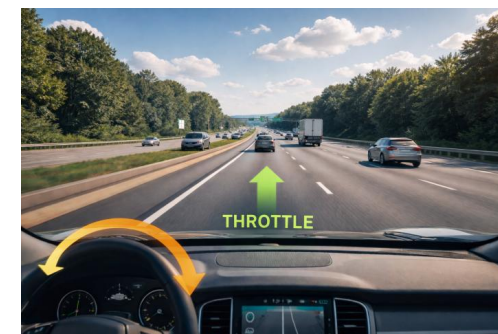
encoder
(e.g., ViT, ResNet)



action distribution
parameters



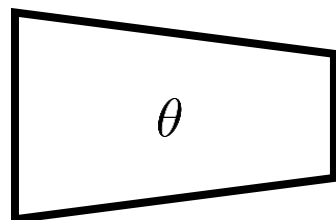
sample



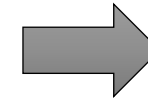
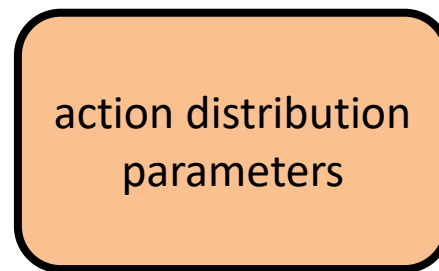
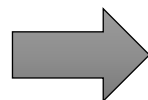
$\mathbf{a}_t$



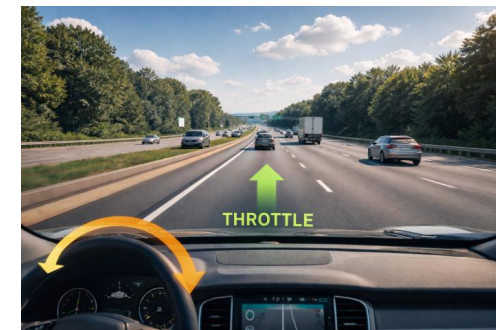
$\mathbf{o}_t$



encoder
(e.g., ViT, ResNet)

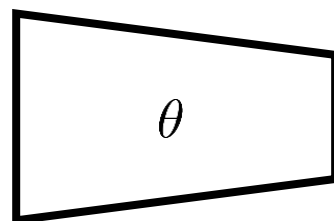


sample



$\mathbf{a}_t$

Discrete actions
(e.g., tokens for LLM,
key presses for Atari games)



output = logits

1-A个动作分布 (未归一化)

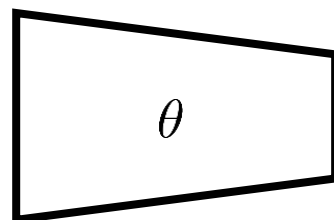
$$\begin{bmatrix} f_1(\mathbf{o}_t) \\ f_2(\mathbf{o}_t) \\ \dots \\ f_A(\mathbf{o}_t) \end{bmatrix}$$

$$p(a_t = 1 | \mathbf{o}_t) = \frac{\exp(f_1(\mathbf{o}_t))}{\sum_{i=1}^A \exp(f_i(\mathbf{o}_t))}$$

用softmax归一化生成action概率分布

We'll learn about other more advanced ways to
represent complex action distributions later!

Continuous actions
(e.g., driving a car)



output = distribution parameters

$$\begin{bmatrix} \mu(\mathbf{o}_t) \\ \Sigma(\mathbf{o}_t) \end{bmatrix}$$

均值, 最可能代表的动作
协方差, 代表预测不确定性

$$p(\mathbf{a}_t | \mathbf{o}_t) = \mathcal{N}(\mathbf{a}_t | \mu(\mathbf{o}_t), \Sigma(\mathbf{o}_t))$$

概率密度函数

$$\log p(\mathbf{a}_t | \mathbf{o}_t) = \|\mathbf{a}_t - \mu(\mathbf{o}_t)\|^2$$

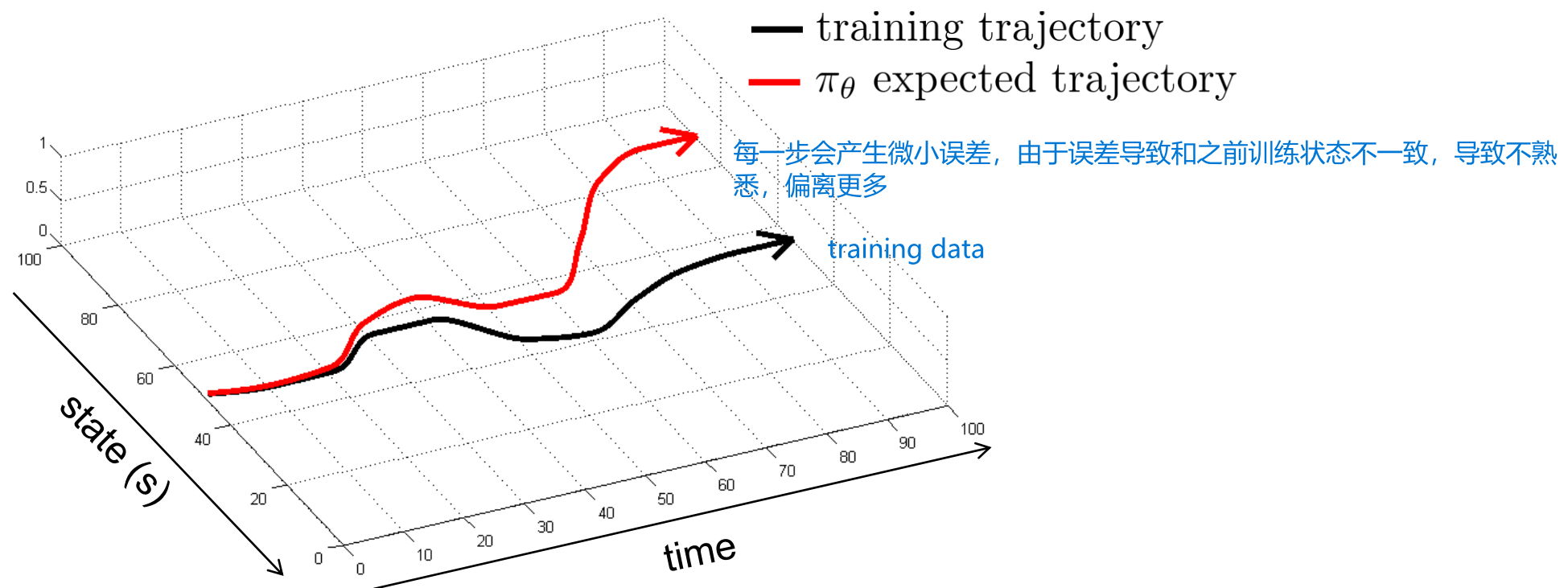
in special case $\Sigma(\mathbf{o}_t) = \mathbf{I}$

Part 3:

Does behavioral cloning work?

Does it work?

No!

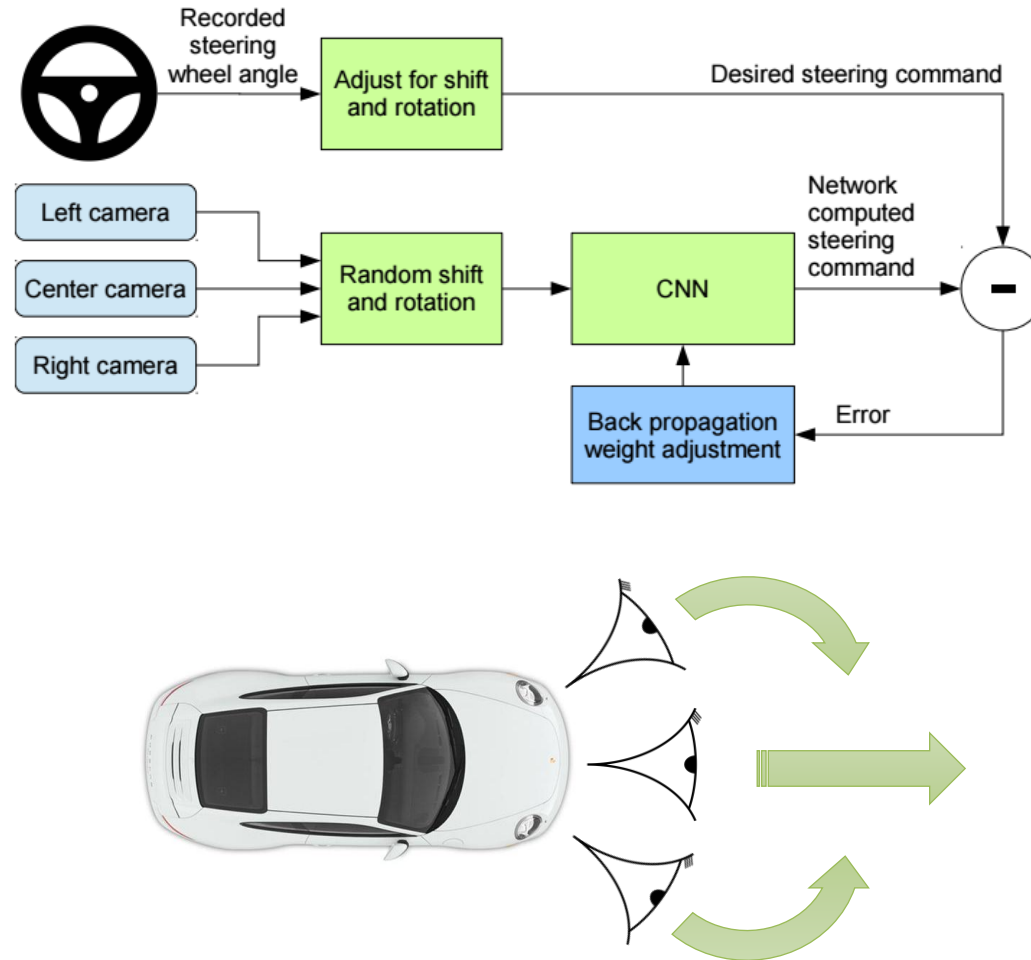


Does it work?

Yes!



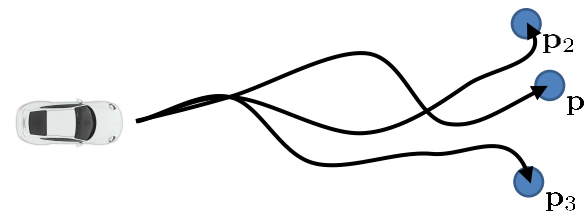
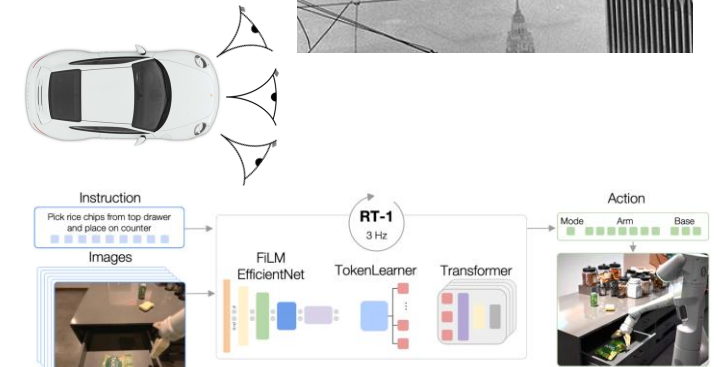
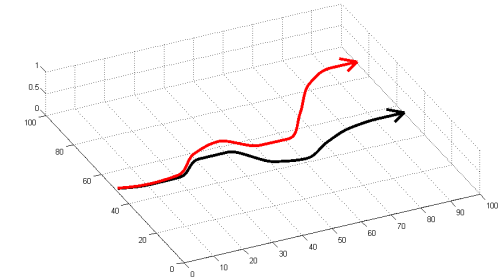
Why did that work?



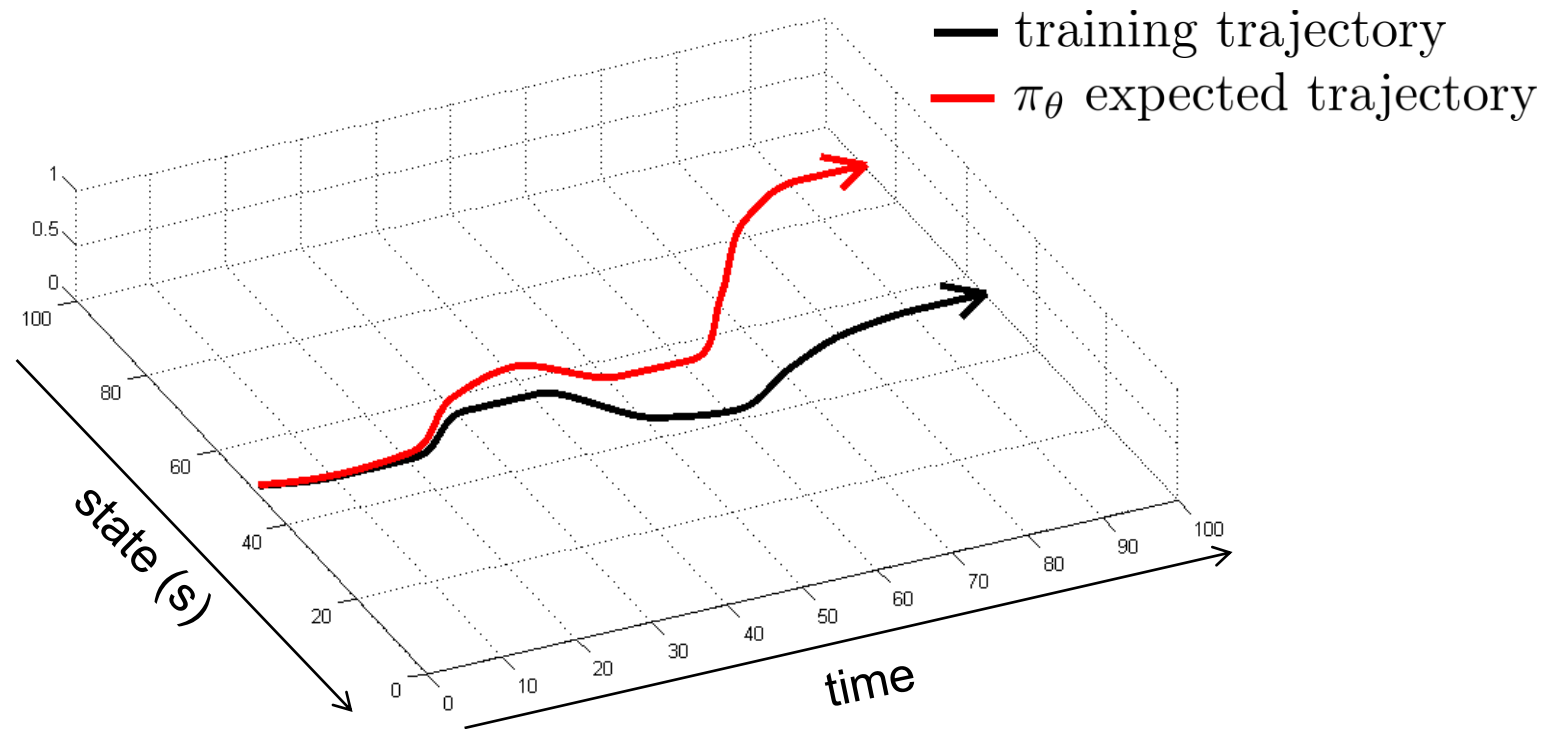
The moral of the story, and a list of ideas

- Imitation learning via behavioral cloning is not guaranteed to work
 - This is **different** from supervised learning
 - The reason: i.i.d. assumption does not hold!
- We can formalize **why** this is and do a bit of theory
- We can address the problem in a few ways:
 - Change the algorithm (DAgger)
 - Use very powerful models that make very few mistakes
 - Be smart about how we collect (and augment) our data
 - Use multi-task learning

独立同分布的假设是不成立的
在RL中，智能体的状态和动作会影响未来的状态和奖励，
所以数据样本之间存在强相关性。
随着策略的更新，智能体探索的状态空间会改变，
导致采样的数据分布不断变化。



Back to this...



Distributional shift

Definition: we train $p_{\theta}(y|\mathbf{x})$ on $\mathbf{x} \sim p_{\text{train}}(\mathbf{x})$

we test $p_{\theta}(y|\mathbf{x})$ on $\mathbf{x} \sim p_{\text{test}}(\mathbf{x})$

$$p_{\text{test}}(\mathbf{x}) \neq p_{\text{train}}(\mathbf{x})$$

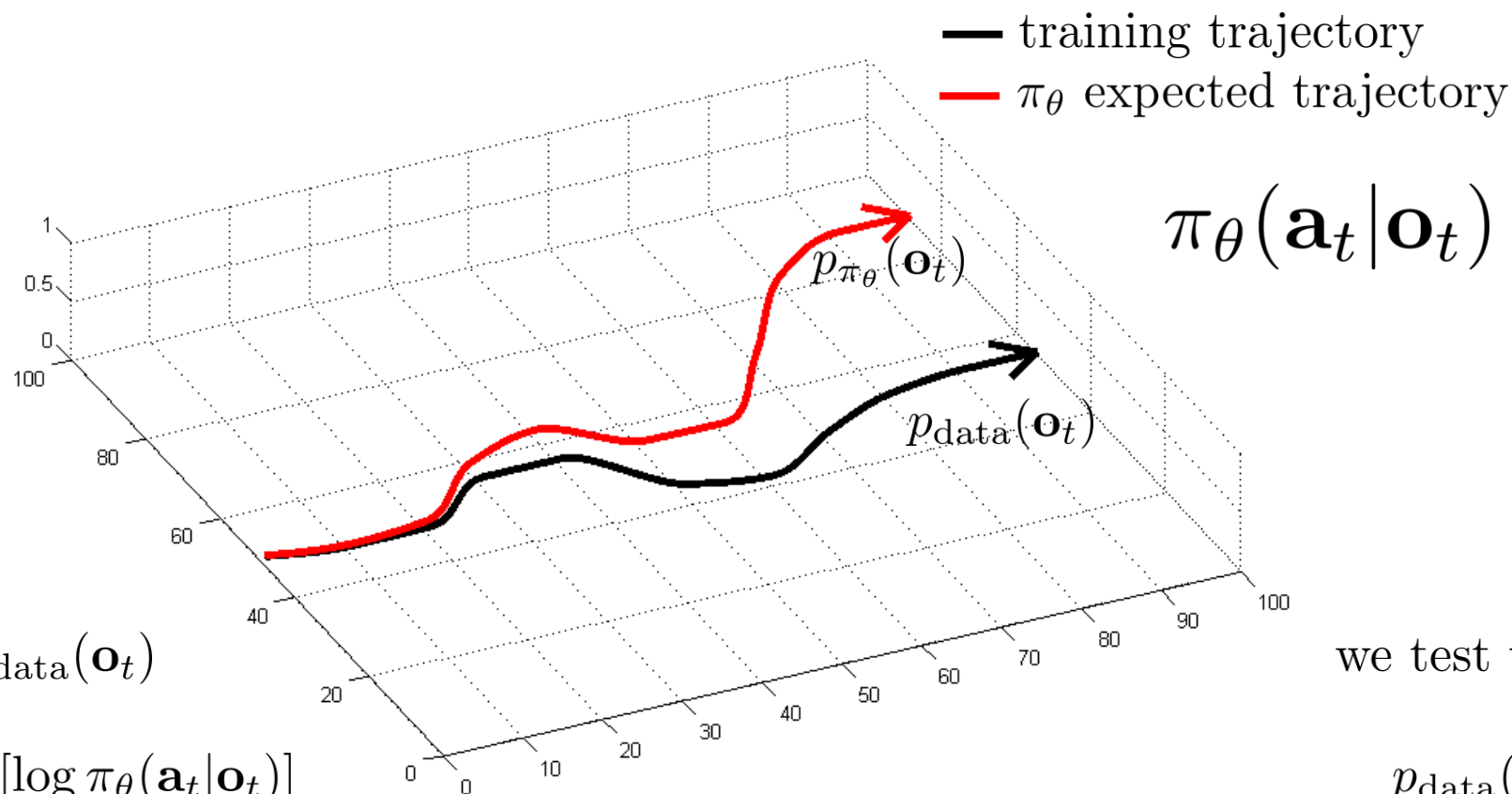
How bad could it be?

Imagine you prepare for a math exam, and
get an exam on ancient Greek literature



Distributional shift with BC

后面没讲



we train under $p_{\text{data}}(\mathbf{o}_t)$

$$\max_{\theta} E_{\mathbf{o}_t \sim p_{\text{data}}(\mathbf{o}_t)} [\log \pi_\theta(\mathbf{a}_t | \mathbf{o}_t)]$$

we test under $p_{\pi_\theta}(\mathbf{o}_t)$

$$p_{\text{data}}(\mathbf{o}_t) \neq p_{\pi_\theta}(\mathbf{o}_t)$$

How bad could it be?



Intermission

Part 4:

Just how bad could it be, really?



Some things we might like to know

Is it worse than supervised learning?

Do the problems vanish with more data?

消失

Does it get worse for some control problems?

Theory can help us answer these questions

What theory **gives** us:

intuition for tradeoffs (data, horizon, etc.)

“how bad can it be” (worst case)

What theory **doesn't** give us:

a practical “proof” that it works (or doesn't)

a way to understand *typical* behavior

Quantifying how bad it could be

Assume the data is produced by a good policy $\pi^*(\mathbf{s}_t)$

(fully observed case for simplicity)

$$c(\mathbf{s}_t, \mathbf{a}_t) = \begin{cases} 0 & \text{if } \mathbf{a}_t = \pi^*(\mathbf{s}_t) \\ 1 & \text{otherwise} \end{cases}$$

incur a cost of 1 if you make a “mistake”

$$\mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)}[c(\mathbf{s}_t, \mathbf{a}_t)] = \sum_{\mathbf{a}_t} \pi_\theta(\mathbf{a}_t | \mathbf{s}_t) c(\mathbf{s}_t, \mathbf{a}_t) = \pi_\theta(\mathbf{a}_t \neq \pi^*(\mathbf{s}_t) | \mathbf{s}_t)$$

“probability of a mistake”

↑
expected value w.r.t. actions sampled from π_θ

Quantifying how bad it could be

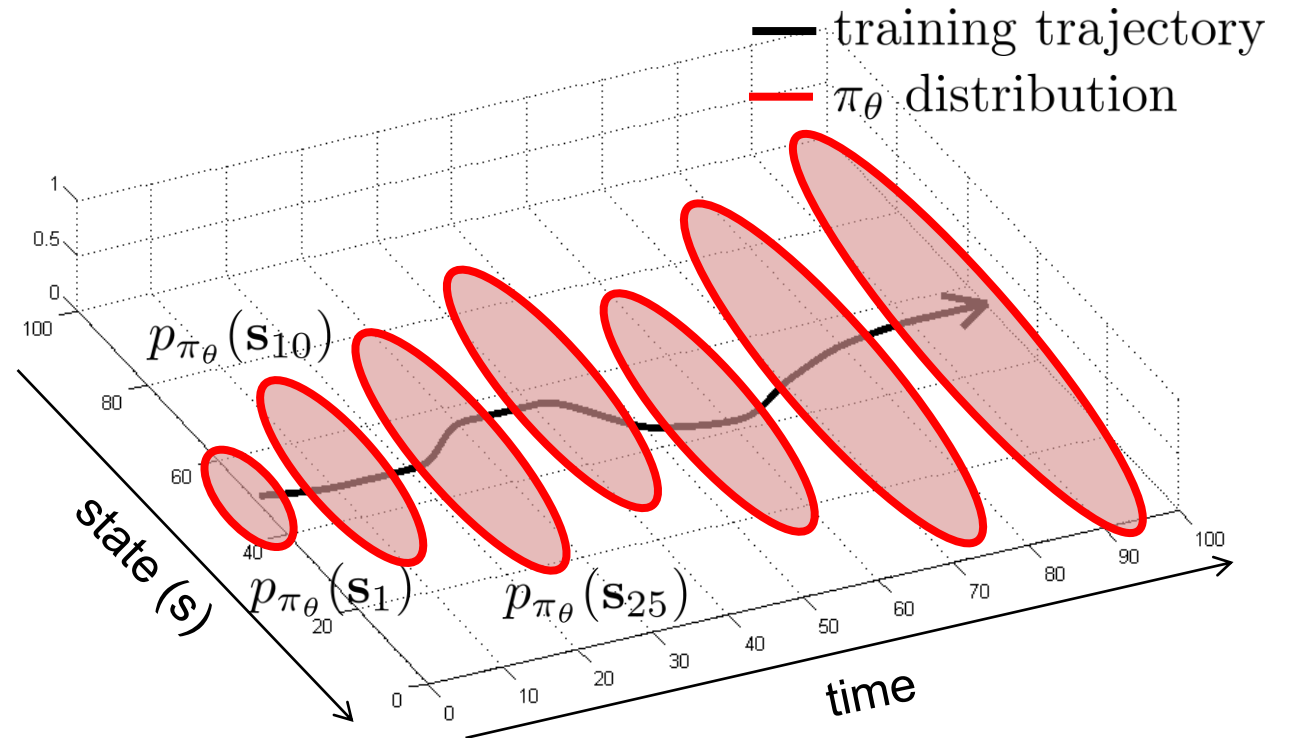
$$\sum_{t=1}^H \mathbb{E}_{\mathbf{a}_t \sim \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t), \mathbf{s}_t \sim p_{\pi_{\theta}}(\mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)]$$

note that this depends
on past actions

This sum corresponds to the
expected total number of mistakes

Question: how does the total number of mistakes scale with the length of the trajectory (the “horizon”)

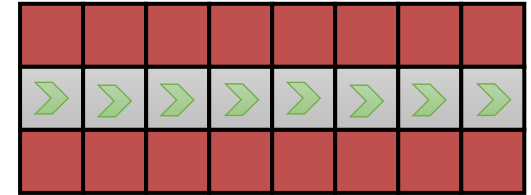
$$c(\mathbf{s}_t, \mathbf{a}_t) = \begin{cases} 0 & \text{if } \mathbf{a}_t = \pi^*(\mathbf{s}_t) \\ 1 & \text{otherwise} \end{cases}$$



A worst-case situation

assume: $\pi_\theta(\mathbf{a} \neq \pi^*(\mathbf{s})|\mathbf{s}) \leq \epsilon$

for all $\mathbf{s} \in \mathcal{D}_{\text{train}}$



$$\sum_{t=1}^H \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t|\mathbf{s}_t), \mathbf{s}_t \sim p_{\pi_\theta}(\mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \leq \epsilon H$$

$$O(\epsilon H^2)$$

H terms, each $O(\epsilon H)$

A more general analysis

assume: $\mathbb{E}_{\mathbf{s}_t \sim p_{\text{train}}(\mathbf{s}_t)} [\pi_{\theta}(\mathbf{a}_t \neq \pi^*(\mathbf{s}_t) | \mathbf{s}_t)] \leq \epsilon$

“if we sample a state from the training distribution, we are unlikely to make a mistake”

A useful property:

$$p_{\pi_{\theta}}(\mathbf{s}_t) = \underbrace{(1 - \epsilon)^t p_{\text{train}}(\mathbf{s}_t)}_{\text{probability we made no mistakes}} + \underbrace{(1 - (1 - \epsilon)^t)}_{\text{some } \textit{other} \text{ distribution}} p_{\text{mistake}}(\mathbf{s}_t)$$

A more general analysis

Total variation divergence $D_{\text{TV}}(p, q) = \frac{1}{2} \sum_x |p(x) - q(x)| \leq 1$

$$p_{\pi_\theta}(\mathbf{s}_t) = (1 - \epsilon)^t p_{\text{train}}(\mathbf{s}_t) + (1 - (1 - \epsilon)^t) p_{\text{mistake}}(\mathbf{s}_t)$$

$$D_{\text{TV}}(p_{\text{train}}, p_{\pi_\theta}) = \underbrace{\frac{1}{2} \sum_{\mathbf{s}_t} |p_{\text{train}}(\mathbf{s}_t) - p_{\pi_\theta}(\mathbf{s}_t)|}_{\substack{\frac{1}{2} \sum_{\mathbf{s}_t} |p_{\text{train}}(\mathbf{s}_t) - (1 - \epsilon)^t p_{\text{train}}(\mathbf{s}_t) - (1 - (1 - \epsilon)^t) p_{\text{mistake}}(\mathbf{s}_t)| \\ \uparrow \\ (1 - \epsilon)^t p_{\text{train}}(\mathbf{s}_t) + (1 - (1 - \epsilon)^t) p_{\text{train}}(\mathbf{s}_t) \\ = (1 - (1 - \epsilon)^t) \frac{1}{2} \sum_{\mathbf{s}_t} |p_{\text{train}}(\mathbf{s}_t) - p_{\text{mistake}}(\mathbf{s}_t)|}}$$

useful identity:

$$(1 - \epsilon)^t \geq 1 - \epsilon t \text{ for } \epsilon \in [0, 1] \qquad \leq (1 - (1 - \epsilon)^t) \leq \epsilon t$$

A more general analysis

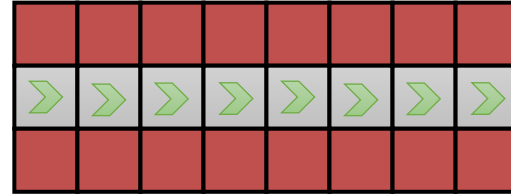
$$\begin{aligned} & \sum_{t=1}^H \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t), \mathbf{s}_t \sim p_{\pi_\theta}(\mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \\ &= \sum_{t=1}^H \sum_{\mathbf{s}_t} p_{\pi_\theta}(\mathbf{s}_t) \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \\ &= \sum_{t=1}^H \sum_{\mathbf{s}_t} (p_{\text{train}}(\mathbf{s}_t) + p_{\pi_\theta}(\mathbf{s}_t) - p_{\text{train}}(\mathbf{s}_t)) \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \\ &\leq \sum_{t=1}^H \sum_{\mathbf{s}_t} (p_{\text{train}}(\mathbf{s}_t) + |p_{\pi_\theta}(\mathbf{s}_t) - p_{\text{train}}(\mathbf{s}_t)|) \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \\ &= \sum_{t=1}^H \left[\sum_{\mathbf{s}_t} p_{\text{train}}(\mathbf{s}_t) \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \right] + \left[\sum_{\mathbf{s}_t} |p_{\pi_\theta}(\mathbf{s}_t) - p_{\text{train}}(\mathbf{s}_t)| \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \right] \end{aligned}$$

A more general analysis

$$\begin{aligned}
 & \sum_{t=1}^H \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t), \mathbf{s}_t \sim p_{\pi_\theta}(\mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \\
 &= \sum_{t=1}^H \left[\underbrace{\sum_{\mathbf{s}_t} p_{\text{train}}(\mathbf{s}_t) \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)]}_{\mathbb{E}_{\mathbf{s}_t \sim p_{\text{train}}(\mathbf{s}_t)} [\pi_\theta(\mathbf{a}_t \neq \pi^*(\mathbf{s}_t) | \mathbf{s}_t)] \leq \epsilon} + \underbrace{\sum_{\mathbf{s}_t} |p_{\pi_\theta}(\mathbf{s}_t) - p_{\text{train}}(\mathbf{s}_t)| \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)]}_{= 2D_{\text{TV}}(p_{\pi_\theta}, p_{\text{train}}) \leq 2\epsilon t} \underbrace{\leq 1} \right] \\
 &\leq \sum_{t=1}^H \epsilon + 2\epsilon t \quad O(\epsilon H^2) \quad \text{模仿学习误差会随着时间步长H二次幂增长}
 \end{aligned}$$

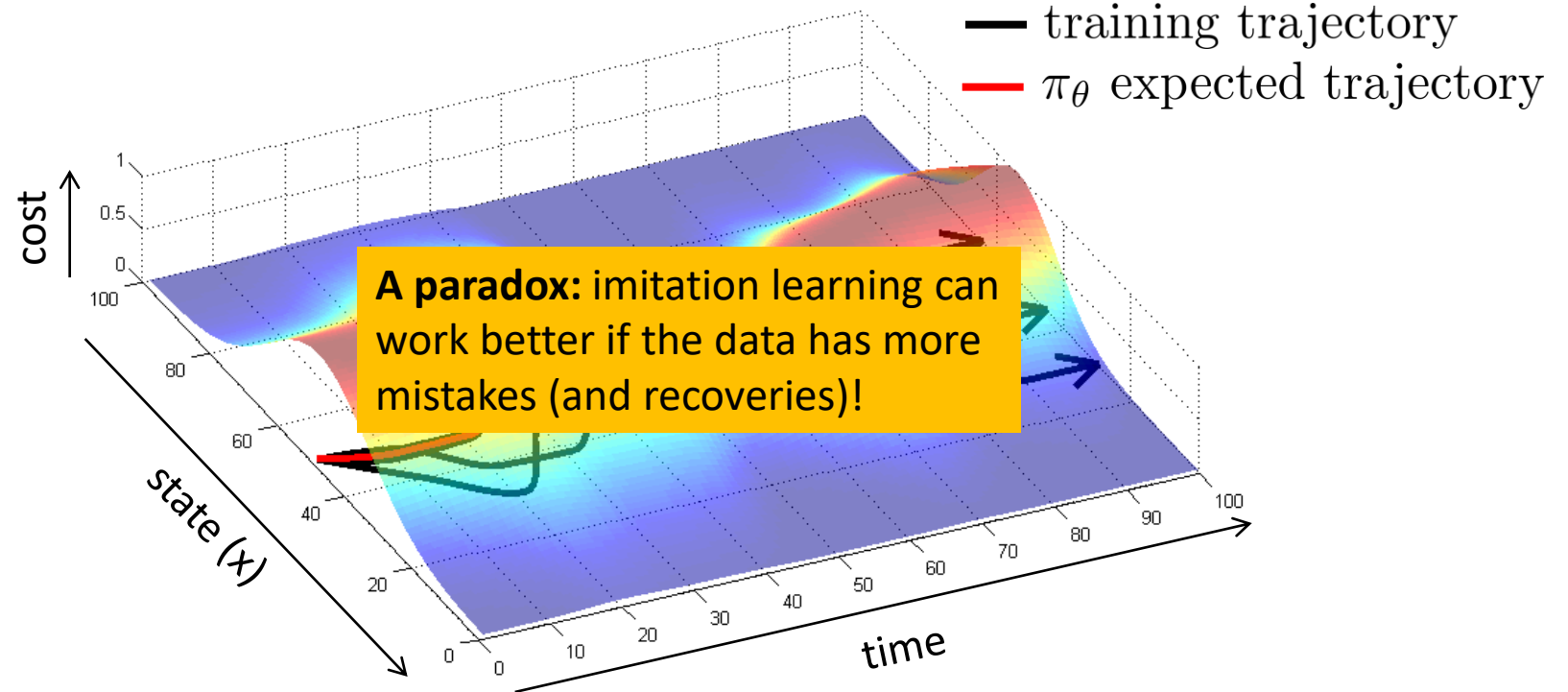
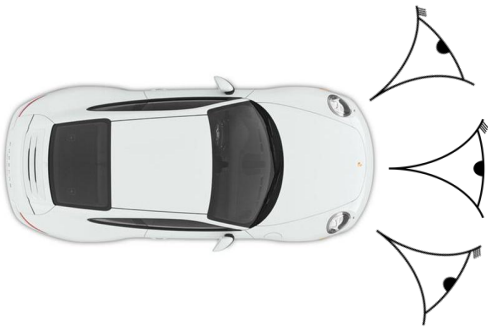
This is bad because error increases *quadratically* with horizon

Why is this rather **pessimistic**?



In reality, we can often **recover** from mistakes

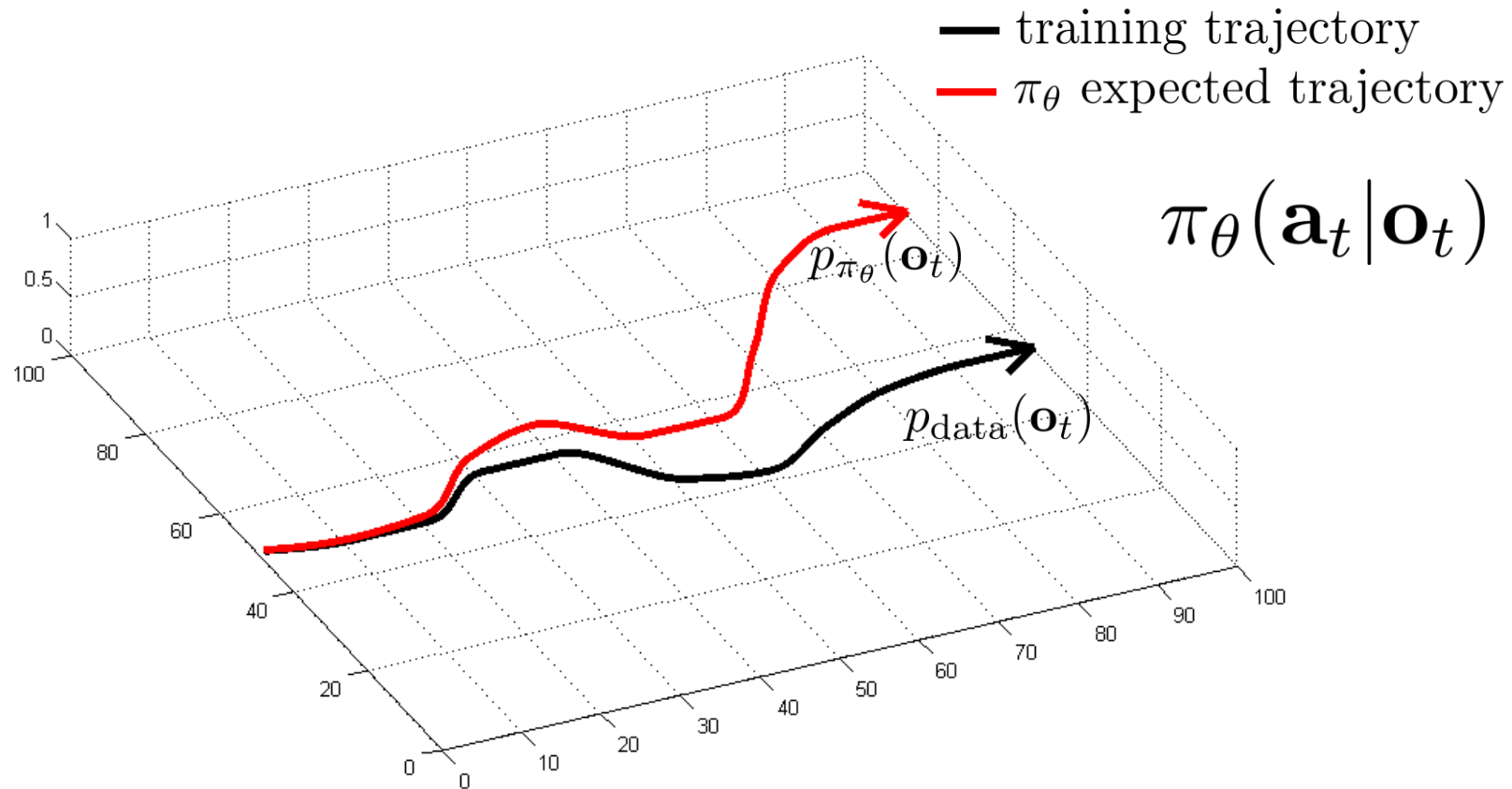
But that doesn't mean that **imitation learning** will allow us to learn how to do that!



Part 5:

Can we do *better* than behavioral cloning?

What can we do about it?



can we make $p_{\text{data}}(\mathbf{o}_t) = p_{\pi_\theta}(\mathbf{o}_t)$?

Fixing distributional shift

can we make $p_{\text{data}}(\mathbf{o}_t) = p_{\pi_\theta}(\mathbf{o}_t)$?

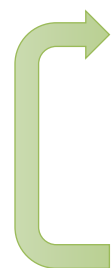
idea: instead of being clever about $p_{\pi_\theta}(\mathbf{o}_t)$, be clever about $p_{\text{data}}(\mathbf{o}_t)$!

DAgger: Dataset Aggregation 数据集聚合而不是模型聚合

goal: collect training data from $p_{\pi_\theta}(\mathbf{o}_t)$ instead of $p_{\text{data}}(\mathbf{o}_t)$

how? just run $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$

but need labels $\mathbf{a}_t$!

- 
1. train $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ from human data $\mathcal{D} = \{\mathbf{o}_1, \mathbf{a}_1, \dots, \mathbf{o}_N, \mathbf{a}_N\}$ 传统监督学习
 2. run $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ to get dataset $\mathcal{D}_\pi = \{\mathbf{o}_1, \dots, \mathbf{o}_M\}$ 用监督学习之后的模型跑一组观测值
 3. Ask human to label $\mathcal{D}_\pi$ with actions $\mathbf{a}_t$ 人工标注观测值应该对应的action
 4. Aggregate: $\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{D}_\pi$ 聚合原有数据集和新标注的数据集

Dagger Example

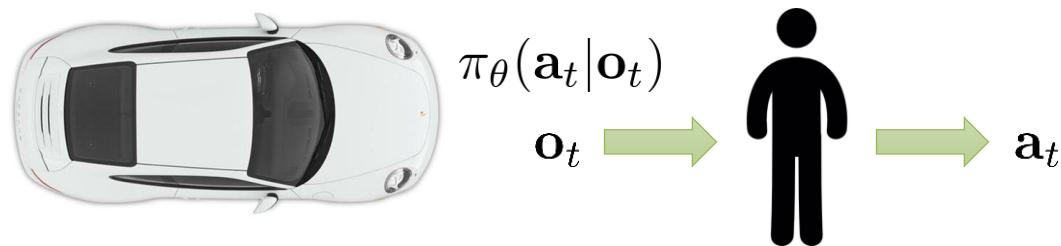


What's the problem?

1. train $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ from human data $\mathcal{D} = \{\mathbf{o}_1, \mathbf{a}_1, \dots, \mathbf{o}_N, \mathbf{a}_N\}$
2. run $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ to get dataset $\mathcal{D}_\pi = \{\mathbf{o}_1, \dots, \mathbf{o}_M\}$
3. Ask human to label $\mathcal{D}_\pi$ with actions $\mathbf{a}_t$
4. Aggregate: $\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{D}_\pi$

这种方法不常用的原因:

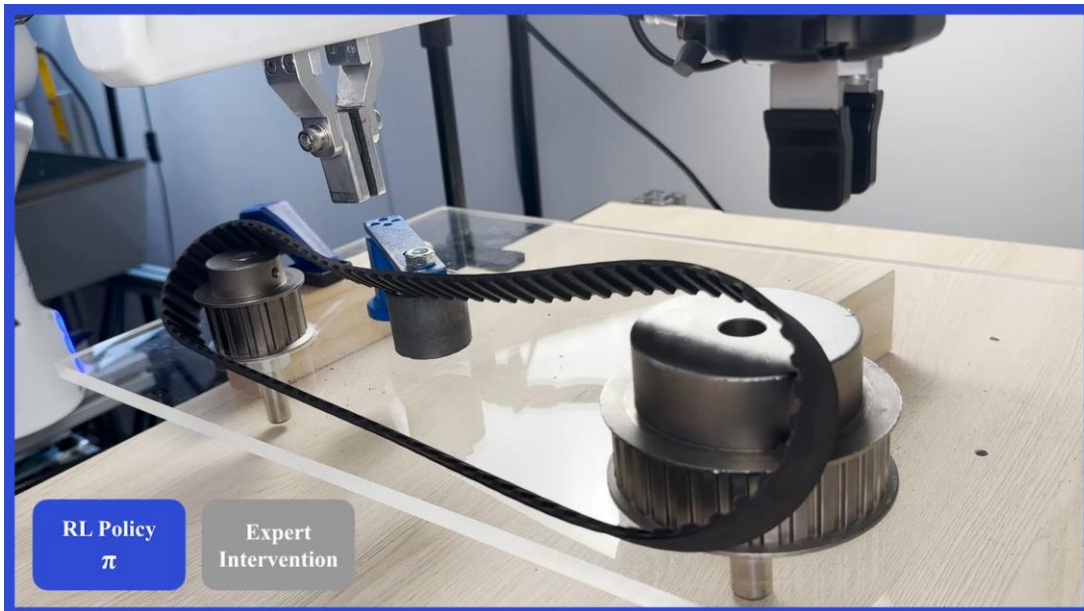
问题出在第三步, 人工标注非常繁重, 而且没有反馈的标注决策是人类非常不适应的



A common variant of DAgger

1. train $\pi_{\theta}(\mathbf{a}_t|\mathbf{o}_t)$ from human data $\mathcal{D} = \{\mathbf{o}_1, \mathbf{a}_1, \dots, \mathbf{o}_N, \mathbf{a}_N\}$
2. run $\pi_{\theta}(\mathbf{a}_t|\mathbf{o}_t)$
3. ask human to *take over* at some time step t
4. store all $(\mathbf{o}_t, \mathbf{u}_t)$ examples from human intervention

Is this still guaranteed to fix distributional shift?



Recap

- Imitation learning via behavioral cloning is not guaranteed to work
 - This is **different** from supervised learning
 - The reason: i.i.d. assumption does not hold!
- We can formalize **why** this is and do a bit of theory
- We can address the problem in a few ways:
 - Change the algorithm (DAgger)
 - Use very powerful models that make very few mistakes
 - Be smart about how we collect (and augment) our data
 - Use multi-task learning

